LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: Structure of the Atom

Sub-atomic particles

Particle	Charge	Mass	Discovered by
Electron	−1	about 1/2000 of a proton	J. J. Thomson
Proton	+1	1 u	E. Goldstein (canal rays)
Neutron	0	about 1 u	J. Chadwick

Models of the atom

- **Thomson:** a positive sphere with electrons embedded in it, like seeds in a watermelon. Atom is neutral overall.
- **Rutherford (α-particle scattering on gold foil):** most α-particles passed straight through, so most of the atom is **empty space**; a few were deflected and very few bounced back, so the positive charge and mass are concentrated in a tiny **nucleus**.
- **Drawback of Rutherford's model:** a revolving electron should lose energy and fall into the nucleus, so the atom would be unstable.
- **Bohr:** electrons revolve in certain discrete orbits (shells K, L, M, N) and do not radiate energy while in them.

Atomic number Z = number of protons · Mass number A = protons + neutrons · Neutrons = A − Z
Maximum electrons in a shell = $2n^2$: K = 2, L = 8, M = 18 (but the outermost shell holds at most 8)
Valency = electrons gained, lost or shared to complete the octet, e.g. Na (2, 8, 1): 1; Cl (2, 8, 7): 1; O (2, 6): 2

Isotopes and isobars

- **Isotopes:** same atomic number, different mass number, e.g. hydrogen-1, 2, 3; carbon-12 and carbon-14. Uses: uranium-235 as nuclear fuel, cobalt-60 to treat cancer, iodine-131 to treat goitre.
- **Isobars:** same mass number, different atomic number, e.g. calcium-40 and argon-40.

Quick check

1. Write the electronic configuration of sulphur (Z = 16) and give its valency.

2. An atom has 11 protons and 12 neutrons. Find its atomic number and mass number.